INTRODUCTION TO FEATURES IN THE STELLAR CYBER USER INTERFACE

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L00 - Introduction and Agenda

L00A-PRES: Intro to Features in SC UI (01m:38s)

00:00-01:00
Welcome to the *Introduction to the Stellar Cyber User Interface* module of the *SOC Analyst* course.In this module, we'll take a look at the features and functionalities of the user interface and how to find what you're looking for in order to complete investigations. We won't be looking at the user interface in terms of left to right or top to bottom, but rather in terms of the important features that you'll need to complete investigations with the outcome you seek to achieve. We'll get started by looking at Case views. *Case Views* explores the geography of working with Cases in the user interface. We'll take a look at the **Top Cases** view, the **Case Table** view, and the **Kill Chain** view, and show how you use them to complete investigations. We'll have an overview of the user interface as it pertains to triaging and investigating Cases. We'll give a demo of the Case investigation tools found in the UI. We'll take a

look at Alerts in the user interface and how to interact with Alerts.

01:00-01:38
We'll explore the Alerts views presented in the UI so we'll look at **Alerts Table** view and the **Investigate | Threat Hunting** view. We'll give a demo of how to interact with Alerts. Next, we'll take a look at threat hunting. This will explore threat hunting tools in the UI in order to find the unknown threats that reside on your network. We'll move on to user and asset analytics to show how to add valuable context to your investigations and we'll give a demo of these analytics. Now that we have an idea of what's going to be presented in this module, let's move on to the next lesson and get started.

L01 - CASE Management

L01A-PRES: Case Views (11m:57s)

00:00-01:00
Welcome to the *Introduction to the Stellar Cyber User Interface* module, lesson *Case Views*. In this lesson, we'll be talking about Case views in the user interface. This is one of the most important sets of views for an analyst securing a network with Stellar Cyber. We'll discuss why in this module. Later, we'll go deeper into the subject in the *Cases and Case Management* module of the *SOC Analyst* course. We'll explore how automatic scoring of Cases helps push the riskiest Cases to the top of analysts' attention and prioritizes triage and investigation. The **Top Cases** view is one of the views that helps analysts see the highest-level threats first. The **Cases** view shows all Cases generated in a specified time frame, creating visibility into all Cases and helping to set workflow expectations for any given day. Throughout the lesson, we'll demonstrate how to navigate the **Case** views to achieve desired security outcomes and get to the places we need to get to to perform effective threat triage and investigation.

01:00-02:00
What is a Case? Let's talk a little bit about Cases and why they are important. Stellar Cyber's correlation of Alerts and security events into Cases is an efficiency force multiplier for security investigations. Having related events tell the story of what is happening in a breach or security incident greatly simplifies an analyst's workflow. Sometimes analyzing individual Alerts is an effective strategy, but being able to see correlated events and having the ability to give a status to all Alerts and events in the Case at once, saves a substantial amount of analysis and response time. It also provides a more comprehensive picture of the situation and prevents analysts from having to manually correlate events themselves. It keeps threats from being misunderstood by painting a complete picture of what is occurring on your network.

02:00-03:00
The way this picture is painted is massively important to being able to reduce mean time to response. Here we can almost instantly tell what is happening in this dangerous security incident. A phishing URL was accessed from this host. Threat intelligence Enrichment has warned us of a malicious URL. Here we see that an internal Trojan was deployed, as seen by system rules. Clearly this is not positive

behavior on this host. Next, we see a remote desktop protocol session is opened, which means access to the system remotely for an attacker. The RDP session is hijacked, giving the attacker remote control over the session. Then we see PowerShell remote access, which allows the attacker to execute commands on the network. That's just the first few Alerts we've seen in this Case, and we already know what is happening.

03:00-04:00
We can take instant action against these attempts to harm our network. Selecting the drop-down menus gives us further details about the Alerts. Allowing us to understand the activity even better.

So how do Cases get built? Stellar Cyber employs advanced analytics, machine learning, and integration capabilities to correlate security events and create Cases. The process generally involves several key steps, which are typical for advanced security platforms that utilize big data analytics and artificial intelligence to enhance cybersecurity operations. While specific implementations can vary, the overarching mechanism follows a broadly similar pattern.

Data aggregation. Stellar Cyber collects data from various sources across the network, including endpoints, network devices, cloud environments, and other security tools. This data includes logs, flow packets, Alerts, and others.

04:00-05:00
Normalization and enrichment. The collected data is then normalized, meaning it is transformed into a consistent format for analysis. Additionally, the data is ~~may be~~ [MOU1] [PW2] Enriched with external threat intelligence to provide context, such as identifying known malicious IP addresses, domains, and file hashes.

Detection and analytics. Using machine-learning algorithms and behavioral analytics, Stellar Cyber analyzes the normalized and Enriched data to detect anomalies, patterns, and signs of security threats. This involves comparing activities against baselines of normal behavior and identifying deviations that may indicate a security issue.

Event correlation. Once threats are detected, Stellar Cyber correlates related events to construct a coherent narrative of an attack or security incident. This correlation is based on various factors such as time frames, entities involved, like IP addresses, user IDs, and device IDs, and the nature of the activities. The goal is to link disparate events that may seem unrelated at first, but together form a sophisticated cyber attack.

05:00-06:00
Case creation. From the correlated events, Stellar Cyber automatically creates Cases. Each Case represents a potential security incident that requires investigation. The platform prioritizes Cases based on severity, impact, and other criteria to help security analysts focus on the most critical issues first.

Incident investigation and response. Security analysts review the Cases, investigate the underlying events, and determine the appropriate response. Stellar Cyber's platform provides tools for in-depth analysis such as the ability to drill down into raw data, use forensic capabilities, and leverage automation for rapid response actions like isolating affected systems or blocking malicious traffic. Stellar Cyber integrates with a wide range of security tools and platforms, allowing it to serve as a central hub for security operations.

06:00-07:00
This integration capability is crucial for correlating events across diverse data sources and security layers, enabling a comprehensive view of security posture and more effective incident detection and response.

Case views, why do they matter? The Top Cases view appears on the homepage and alerts analysts to the most riskiest events on the network so that you can prioritize investigations accordingly. The Cases view shows all Cases generated in the time frame and allows analysts to assess what their workflow for the day looks like. Both of these views allow linking into the Cases generated by the system. These Cases are the same from both views but appear in a different way. So how do Cases get scored? Here's how it works.

Stellar Cyber generates scores on Cases by leveraging machine learning to correlate disparate Alerts into coalesced Cases. A Case is a grouping of potentially related Alerts bundled into a single data structure...

07:00-08:00
...which provides a holistic context for analysts. This approach allows security analysts to assess whether a Case represents a real attack, true high-risk behavior, or event connections with a low level of security significance.

The correlation algorithm works by associating an Alert with a Case in real time, grouping and correlating the Alert with an existing Case if there's a strong connection or creating a new Case if no strong connection exists. This process continues until a Case is closed by a user or no new Alert is associated with that Case for a certain period of time.

The score of a Case is dynamically updated based on its criticality with the scoring system designed to reflect the severity of the situation. The score increases with the number of different Alert types associated with the Case, affected by the fidelity, or the relative confidence in the analysis, and severity, or the importance of the category of the event of the associated Alerts.

08:00-09:00
Fidelity reflects confidence that a malicious event was correctly observed, while severity indicates the potential danger of the event. Stellar Cyber calculates an event score for each different Alert type associated with the Case by summing the maximum Alert score, severity, and fidelity for all individual Alerts of a given type. The final score of the Case is then derived by normalizing the total score from all event scores.

In the Stellar Cyber architecture, sensors, connectors, and machine-learning algorithms play crucial roles in detecting and reporting Cases. Sensors capture and send observed events to the system, while connectors collect information from the external data sources. The machine-learning component examines the data stored in the Data Lake for evidence of security breaches, generating Alerts that contribute to Case formation. The system's ability to correlate Alerts into Cases and assign scores based on the criticality of these Cases is central to its operation, aiding in the prioritization and response to potential security threats.

09:00-10:00
The Top Cases workflow is on the homepage, showing the highest risks first. Ranking by score allows for triage of the riskiest events quickly. These Cases point at the same events that appear in the Cases view we'll discuss next but just gets you to them in a prioritized manner. Cases can be sorted by score in the Cases view too, but this view is for expediency and efficiency in triage and investigation workflows. When expanding the Cases, a more detailed view can be seen by the analyst, helping them to decide if they want to click through the top Case link into the full Case for investigation. The Cases workflow view presents all Cases created in the time frame set by the filter on the left. Selecting the link highlighted in this image takes you to the full list of Cases. The Cases view looks like this in the UI.

10:00-11:00
You can see that all Cases appear in this view. This allows analysts to assess how many Cases need evaluation that day, very quickly. It allows analysts or SOC leadership to set priorities for the day. The Case table can be sorted, filtered, and edited to show the fields of greatest interest first in the fields. Scroll to the right within the table to see all fields chosen. Viewing all of the Cases created in the filtered time range allows for a quick assessment of how much work there is to be accomplished on a given day.

Let's take a moment to check your knowledge.

True or False? The Cases view only shows the most critical Cases, not all Cases.

The correct answer is False. All Cases are shown in the Cases view, enabling analysts to quickly assess how many Cases need evaluation that day.

11:00-11:57
How does Stellar Cyber correlate Alerts into Cases?

1. By using a random number generator
2. By employing advanced analytics and machine learning
3. Solely based on the severity of the Alerts
4. Based on the age of the Alerts

The correct answer is B. Stellar Cyber employs advanced analytics, machine learning, and integration capabilities to correlate security events and create Cases.

What does the score of a Case in Stellar Cyber reflect?

1. The severity and criticality of the situation
2. The age of the Case
3. The number of analysts working on the Case
4. None of the above

The correct answer is A. The score of a Case is dynamically updated based on its criticality with the scoring system designed to reflect the severity of the situation.

The *Case Views* lesson is complete. You should now be familiar with Case views, one of the most important views for analysts securing a network in the Stellar Cyber user interface. Thank you.

L01B-DEMO: UI Overview with Cases (15m:14s)

00:00-01:00
The next lesson in the *Introduction to Features in the Stellar Cyber User Interface* training module is a *Demonstration of the UI Overview with Cases*.
In this demonstration, we're going to be taking a look at the user interface, but in the very specific context of how you interact with Cases. The reason we're putting the emphasis on Cases in the user interface is because of the benefits they bestow upon analysis. So looking at Cases as a whole, and Cases are a set of Alerts and events that are "stitched" together by machine learning and analytics in Stellar Cyber to give a picture of what's happening in the event itself. Not just looking at the single Alerts, one at a time. Because you can't really get the context that you need to take immediate action from one single Alert, most of the time. So I've worked with SOCs before.

01:00-02:00
I've done the analysis in a single security platform, and I've done it in aggregated security platforms, and the difference is night and day. When you have a collated, chronological view of all of the events that are occurring together to the same users, to the same assets, on the same entities, it really paints a picture of what is happening on your system. And the term that I typically use is when you're looking at a single vendor platform, it's almost like looking at your network through a coffee straw. You just see a tiny little picture of your network and it doesn't give the whole picture of what you really need to know to take action on a specific set of events.

Another benefit that this methodology confers is that you are able to give a status or disposition to lots of events at the same time.

02:00-03:00
You might be able to knock out 50 Alerts, if it's a false positive, or if it's something you want to put a pin in, or send to someone else, you can give that status to all of the events at one time, and then you have basically reduced your Alerts that you have to triage significantly just by looking at the Cases, because the Cases tell you a story about of what is happening. The machine learning and the analytics stitch together events that it sees some connection to.

If an RDP session was opened, and then there was some sort of external trojan deployed, and then there maybe there was some PowerShell scripting, all those things in isolation do tell you something, but they don't really tell you the chronological picture of how the attack is unfolding. This is why we're

starting with Cases. They just add so much value to your analysis. It takes so much of a load off of analysts from having to do that manual correlation.

03:00-04:00
It eliminates a lot of the opportunity for analysts to get things wrong. Or to give a status of dismissed or get rid of the Alert just because it's not fully understood in the context in which it happened. Having said all that, we're going to jump into the demonstration right now to get some hands-on time in the product itself. So, here we go.

Alright. Let's hop into this demo about the user interface but related to working with Cases in the user interface. We mentioned in the PowerPoint presentation that the two principal ways that you're going to interact with Cases from the homepage will be this **Top Cases** view, and also from this **Cases** view. Those will be the two ways that you interact most commonly and generally you will start by being in the homepage. If you're looking to triage the highest score incidence, you can start with **Top Cases**.

04:00-05:00
If you want to look at all the Cases, click here and it will show you every Case that is within the time frame that you have set. So in this case, we're looking at the eighteenth through the twenty-sixth, so quite a few. So those are the two ways that you'll be looking at it. So let's get acquainted with how this works from the home page. So one thing that's very important to know is that you have the option to set filters and here is a set of filters and here are time filters and we'll explore both. You can filter by Alert scores; drag this slider up or down. You can change both ends of the range. You can look for queries that you have prebuilt and saved. Any query that you've made is searchable. If you've saved it, you can search by status. If we're looking for new events or we're looking for things that have been closed by analysts just to make sure they did those, they did their due diligence correctly.

05:00-06:00
Then we have an option to add all kinds of things to the new filter. So this will create its own drop-down menu for each of these. If you want to add asset tags as a query or assignee as a filter, or event tag as a filter, fidelity, an individual sensor, the severity, threat intel, usernames. All those can function as filters for you if you want to just see the things that you precisely are interested in at any given moment. Likewise we'll move from left to right on here. This search allows you to search for anything that you're looking for. In this case, I'm looking here at a Mimikatz credential dump for example, I want to see anything that involves Mimikatz because that's something that I'm interested in finding on my network. If you're not familiar with Mimikatz, it's basically a tool that was built to find vulnerabilities in Windows products,

06:00-07:00
...and it made its way into the hands of hackers and bad actors, and it's used a lot to do credential dumping and various other things on endpoints, which is universally bad. We're looking for Mimikatz and obviously we see some things that contain Mimikatz here, but some things that don't. So let's see. Why did this come up with Mimikatz. So if we do a search on here, we see that this Mimikatz.exe was used in this event. So it may not be in the title of the Case, but it will be somewhere located in the Case. So we'll get rid of that because one of the reasons we're going over filters and searching first is because if you leave these things on, they go across pages, and you might be missing things that you think you should be seeing but it's because your filter is wrong. And because you've set a filter and just forgot to get rid of it. A lot of times that happens with time ranges. You might see different things in your Top Cases view that you don't see in your Cases view and it's because you have something filtered.

07:00-08:00

Looking at the time filters, we have three options here. One of them is to use an absolute date. So that's a from and a to date. You can set that however you want. And then a **daily** is going to be for about the past twenty-four hours generally. And then **relative** is the one that you will probably use in daily operations because you want to see the Alerts and Cases that you have missed while you have been gone. So if that's overnight, if that's over a weekend, if, you know, maybe your analyst is out for a week, and you don't see things for a week. So this is typically the way that you will set your time range. Okay. Moving back to the Top Cases. What we can see if we expand all of these is that what you get with the Top Cases are the top scores for Cases in the time range that you've set. And this is a really efficient way to sort of find what your daily workflow is going to look like. These are my top Cases. They're probably the most important ones that I need analysts to look at, so let's just triage these here.

08:00-09:00
Then if we need to drill down on them, we can either click into them or we can go to the Cases View up here and look at the whole thing. Here we have just a certain set of key fields that you might be using, including the score obviously, the time stamp when it started, the type of Case that it actually is. What this does is the highest-ranking Alerts, the highest scoring Alerts will typically generate these titles. So there may be more going on than just OT exploitation here, but this is the one that generated the most score, so that's what it gives you to give you a rule of thumb to say this is kind of what I'm going to be interested in within this Case. So here we have a view to our tenant. We can look at our status, the severity, we can look at who this Case has been assigned to if anyone has, the number of Alerts, the last time that it was modified, the time that the Case was created, and then the Case creator.

09:00-10:00
Most of the time in the real world it's going to be your creator is going to be the system. But bear in mind that analysts can create Cases on their own. They can do that manually. If that's the case, you might have someone's name here under creator. We can expand one at a time or we can expand them all to take a look at them. It's up to you. So next we'll take a look at the **Case Status.** So this is a demo environment right here, so we don't have any Cases that are closed in your environment there probably will be some. And then we have critical Cases here. That gives us an idea how much work do we have to do and how busy are we going to be today. Let's take a look at the Cases view. This is a really important view, and you have the same filtering ability with a little bit more, tailorability here. You can adjust your score. You can look at only certain severities. You can look at assignees and filter by assignees.

10:00-11:00
You can change the time range that you have for these Alerts. This one is going from the eighteenth to the twenty sixth. So that's the reason that we have quite a few of them. And then by default your status is going to show escalated, new, and in progress, which are the ones that you're going to be operating in on a daily basis most days, but you have the option to change these. You can go back and look at the Resolved cases and look at the Canceled Cases to make sure that no one's canceling or resolving things that should not be resolved. Also if you want to use tags this will be covered in another lesson, but if you want to leverage tags to get you to the specific types of activity that you want to see, you're welcome to do that. And just as on the homepage we have a whole bunch of other fields that we can choose to filter on here. So modified by, tenant ID, who created it, the Case name, Case ID, anything that you would like to add to this you can add, and it will become its own drop down.

11:00-12:00

Alright. We'll notice in this view as well, we do have a summation of how many Cases by severity. And if this were a real environment you would have multiple severities here. And then Cases by status, you'll have more statuses than this because this is a demo environment. And then Cases by assignee. So you can see who's working on what at a view individualization, super easy and you can tell what's going on. So one important thing that I should mention is that the Top Cases here, depending on the time range that you've set, this is relative this past week. These Cases are the same Cases that appear here. You may see some disparity between the two, but it's probably because of the filter that you've got set. If the filter is set for a time range or is an accidentally left on filter.

12:00-13:00
You could have different data here, but in theory it should be the same if the filters are the same these are the same events. So there will be one of these in here and we'll take a look at that and see if we can verify that. So ninety-nine point nine Mimikatz. We'll take a look at that. We'll see if we can find it here...Let's see Ninety-nine Mimikatz there's two. So there's probably a little bit of a disparity between the time range we've got selected but we can see that we can find it if we want it. In the Cases view, one of the things that is really useful to know and to use in the course of your investigations is that you can put whatever fields you want into this table. And when you scroll to the right you can see all sorts of additional information that you can use. You can add whichever fields you'd like. This is a handy way to just add the things that you want to see.

13:00-14:00
For example, if you wanted to see tags you could add tags as a field, and we'll do that. So we don't have any tags here, but we potentially could in a real environment. So my last action collapsed this table down a little bit. We can modify that. We can click and drag these open to make these wider, or if we want to, we can rearrange the fields altogether. Severity and score I'd rather see severity first I can just click and drop it, or if I want to see the creator first before I even see the Case name, I can arrange it like that. It's completely up to you how you want to arrange these things whatever works best for your workflow. You've got that flexibility that you can edit these things just like you want to. Here if you'd like to pin the Case name column or any column for that matter, you can pin it to the left and it stays there. So now you're scrolling in your table, but this Case name never goes away. In addition to being able to choose your columns, you can also remove a column.

14:00-15:14
You can **Autosize All Columns**, which is useful if like in this case, the name of the Case is cut off. If you autosize, it will show you all of the information that is in a given field. And then you can reset all columns and it resets it back to default, which is not the most useful thing to have. These two are pinned at this point, which you may not want. Also you can export all this material once you get it in the format that you want to a CSV to share, or, as we'll see, we can upload it to the Evidence Locker later in this lesson. So in the next lesson, we are going to talk about the nuts and bolts in the UI and we will see lots of good information in the next part of this lesson.

The lesson, *Demonstration One, UI Overview with Cases*, is complete. You should now have familiarity with the Stellar Cyber User Interface with Cases.

L01C-DEMO: Example of Case Investigation Tools (17m:03s)

00:00-01:00
The next lesson in the *Introduction to the Stellar Cyber User Interface* module is Demonstration 2 – Example Investigation of a Case with Tools in the Stellar Cyber UI. Demonstration 2 - Example Investigation of a Case with Tools in the Stellar Cyber UI. In the previous lesson we took a look at how to think about Cases and a little bit about the navigation of Cases within the UI. In this lesson, we're

going to take a look at how to actually do investigations in the tools that are provided by Steller Cyber in its user interface. So let's get going.

Alright. So in part one of the Case views demo we saw how to navigate Cases within the user interface via the **Top Cases** view, and the **Cases** view. And we learned a little bit about how the Cases work together from these two views. We also learned a little bit about how to organize your table view so that you can see the fields that you want to see and don't see the fields that you don't want to see.

01:00-02:00
In this one, we'll pick back up with the navigation piece a little bit. So we mentioned this before, but this particular Case is going to be found here in **Top Cases**. And it's also going to be found in the **Cases** view as long as your filters are set the same. So we can see here we've got the Cylance Protect 999. It's a score of one hundred. And then if we go to Cases, we can see that it's here too. And it's here because I searched for it here. But if we look for all the Cases, we can find it as well. It's Cylance Protect. It's a score of one hundred. That's the Case that we're looking for. Let's talk about how to get started doing these investigations. So in the case that you want to start doing your triage and investigation from the **Top Cases** view, we'll click into the Cylance Protect Case, and what we get is a whole bunch of information that tells us lots of good things. So first of all, at the top of the view, we have the name of the Case. We have the tenant.

02:00-03:00
We have the score of the Case, which is 100 in this Case. That's pretty high and pretty dangerous. We have the "who." This is a username Adrian on 10.33.4.208. It looks like credential dumping is the type of activity we're looking at, so we know that that's a potentially very risky behavior to have on our network. The "when" is 9:45 on February twenty seventh, the "where" is Los Angeles. This severity was already set and I went ahead and added a status of In Progress. So this was new, but I added In Progress so that everyone knows that I am working on this Case and I signed it to myself. I will do the same with all the Alerts in this Case. I've already done it here but this is a very important first step for you to do is to set the status to in progress, click apply, and then this status is in progress and everyone in your group or if there are external groups that work with these Cases that you don't have a chance to tell that you're actually working in.

03:00-04:00
This tells everyone that this is in progress. Someone is working on it, and they can tell who by looking at the assignee. So now when we go back to the home page we can take a look at this Cylance Protect and if anybody wants to take a look at it they can open it up and see that the status is currently "In Progress" and that the assignee is me or whichever analyst, happens to be doing it. And likewise in the Cases view, you can see that the Cylance Protect is critical and here, this is a good use case, we want to add the status to our view. So status is in progress. This might be a good thing to drag over so that we know immediately that this thing is in progress. As is this one and we don't need to pursue those, but we can take a look and see it who is working on it if we need to. We'll jump back out into the **Top Cases** view and into the Cylance Protect. Next we have the Case Score Breakdown and that gives us a summary of what's going on in this Case. And it starts by telling us the stages of the kill chain that these Alerts are found in. So here we have four kill chain stages. That's an indication

that this is an advanced attack. We've got initial attempts, persistent foothold attempts, exploration, and propagation, so that is a key to us understanding that this attack is fairly well evolved. So we have six initial attempts and those are going to be trying to get into our network somehow. We have four persistent foothold activity events. That means that they are trying to gain a foothold on our network so that should something happen, say we kill a process that they're using, or say we isolate a host. Maybe they have established a foothold elsewhere and this is something that we definitely need to investigate. Exploration is equally dangerous because that is them looking to enumerate things.

05:00-06:00
They might be looking for ways to exploit ports on our network or vulnerabilities on our network, or they might be looking around at different ways that they can exploit things either with a tool or doing it manually. And then propagation is going to be lateral movement essentially. It means that they are trying to move laterally or horizontally on our network and establish a foothold in more places. And what this means is that this is at a stage of dangerous activity where they might be gaining footholds on other entities on other hosts or in other applications, various ways they could propagate themselves to get a foothold and that means that our investigation is going to be much more difficult. Likewise if you sort the Alerts here in the Case these are all the Alerts in the Case. If we sort them chronologically, we can see that they go in a logical order. So the stages of initial attempts so this is a phishing site, more initial attempts, scanner behavior, persistent foothold establishing encrypted command and control...

06:00-07:00
...more command and control, external trojans, and then they're moving on to exploration. So they're scanning ports, which is a very bad activity especially if this user is not some sort of a system admin, but we've already got indication that there are trojans being involved. So we know this is bad behavior. Propagation, they're looking to dump credentials and exploit more accounts, inject processes, and then more propagation. They're trying to escalate privileges, which is very dangerous because should they get admin privileges, they can do a lot more damage than they could with lower privileges. Also within this view we get a really good idea of what's happening and which order it's happening in. And if we want to explore any of these Alerts, we can go click the information field and actions. And what this does is gives us the stage. It tells us what's the activity. It gives us links that we can select to link out into MITRE's ATT&CK page.

07:00-08:00
If we don't know what initial access is, we can click this or if we don't know what phishing is and most analysts are going to know what phishing is but if we want to explore it in greater depth, we can go and find all the research information that we need, which is a really value-adding feature of the user interface. We also have a plain English description of what's going on. So a suspicious visit was observed in encrypted traffic from an internal IP to an external IP, which has a recently registered domain, which is a clue in and of itself, 100relab.com. It indicates a possible phishing site. And we have the fidelity, the severity, the threat intel involved, the status and the assignee so this Alert has not been assigned to someone, but we'll go in and assign all the Alerts to me. And that way, everyone knows that I'm working on it. And if we want to add the assignee here, we can do that as well...

08:00-09:00
...so that people know that it's in progress and Tim is working on it. We can also find the original records here. Take a look at these records here if we want to. We can also delete Alerts. This is a bit of a risky behavior on the part of analysts just because if you delete this row no one else could see it. Unless you're absolutely sure that this was an erroneous assignment of an Alert to a Case, I wouldn't recommend deleting these Alerts because you can't see them and this tells a story. We'll see how it tells a story in the **Analysis** tab and this is really where you get the value out of this product because you have such a detailed view, and we'll take a look at that. The **Analysis** tab is really the most efficient and effective way to triage and investigate Cases.

09:00-10:00
It gives you so many resources of what you need to know is going on in this Case in an organized manner and it allows you to understand what's happening a lot quicker than other views. So here if we take a look at the **Observables** in the **Analysis** tab really quickly, we have this external host; we have nine of them. That's kind of a lot. We'll have to explore those in-depth. And if we want to drill down on those, we can look them up in VirusTotal and tell what VirusTotal tells us about them. Additionally, there's a 360 Panoramic View that we'll explore in a different lesson, but this gives us an insight into what we're looking at here. Bitbucket.org, external IPs, we can drill down on those if we want. As far as hosts go, we've got four internal hosts here that have been touched by this attack. That tells us how advanced the attack is. These processes here we can drill down on these if we want to or look them up because a lot of times these processes will not be evident to an analyst, but they are easy to look up and we can find out in the timeline view what these things are actually doing.

10:00-11:00
And then we have one user here Adrian, and Adrian is on this 10.33.4.208 IP. And we'll look more closely at that. Additionally, there's the **Timeline** view and this is one of the most important views when dealing with Cases because this tells us how many Alerts we have. It's sorted chronologically; we can make sure that it's sorted in order and we can see the narrative of the attack very clearly. So we looked at this in the **Detection** view, but here we've got a possible encrypted phishing site. And if we drop this down, we've got a plain text description of what's happening here. So this suspicious visit from this internal to this external IP at which domain, we quickly can tell what we're dealing with. Likewise we can go into the **Information** tab, and this shows us the same thing we looked at before. But it's an easy view if we want to click out of it, do some research, explore some of the more detailed metadata within the actual Case...

11:00-12:00
...see if it's in other Cases. Take a look at the details of the metadata here. We can use this net flow to create advanced queries if we want to see if this type of attack is happening elsewhere in our network. And then we have actions here that allow us to use API connections to take actions on specific things and we'll look at that in just a minute. So we've got an encrypted phishing site visit. That's dangerous. We've got a private to public exploit anomaly. So this tells us exactly what's happening, a high number of exploit attempts with the IDS signature user agents Microsoft device metadata retrieval client user agent. So this may be some sort of attempt to gather account

information on our network. Then we have scanner behavior, outbound destination country anomalies. Here we have a new host communicating with Ireland and that's not normal for our network because we don't operate there in this situation.

12:00-13:00
And then more public-to-private exploits external user data volume. So this is looking at a comparison of what type of volume usually goes out of this computer, and this is an abnormal amount. Then we have encrypted command and control, which is dangerous. All of this is telling us that this is very dangerous activity. We've got an external trojan. All of this is evident and if we hover over these activities, we can see which account is doing it. This 10.33.4.208 is communicating with this domain 100relab.com. We can see the link between the two entities highlights on each one. So this graphical view is really helpful and it serves as a really useful view to forward to security stakeholders to show them exactly what's happening. If you want to give them the step by step of what's happening, which entities are involved, what steps need to be taken to mitigate this attack, and we could do this directly in the **Response** tab.

13:00-14:00
We'll take a look at it in just a minute, but we want to see this 185.230.63.171 will not cause any risk to our network. If we go ahead and block this IP while we do this investigation. We may eventually want to contain this host, but sometimes you have to do the risk benefit analysis of containing a host if it's a host that's critical to your business operations. You might want to do that research before you block it or contain it or isolate it. We'll go ahead and decide that we want to block this 185.230 IP so that it's not infecting other assets on our network. Here's how we do this. In the **Response** tab we have the actions that are available at the Case level. So there may be available actions at the Alert level. At the Case level we can go ahead and block this IP. We'll pick the IP. The connector is Barracuda. We don't want this thing sending anything into our network. Duration--we will make it for five hours while we do this investigation, and we will **Submit**. I won't submit it here because this is a test environment, but that's how it works. And if we decide that this is dangerous activity enough to disable the user or contain the host or if we have a set of scripts that we want to run to mitigate this type of behavior, we could do all of this from the Case level. The last piece of this Case view's UI demo I'd like to display is the **Case Activity** view and **Evidence Locker** view. These will be covered in much more depth in the Cases and Case Management module of this course, but it does bear on how you transfer information between team members, team units, or maybe even external consumers of this information, or even information security management.

15:00-16:00
Case Activity is basically an audit log of everything that's been done to this Case either manually or automatically, and you can see all the things that have happened that have affected this Case. Additionally, you could add your own comments. Say if you took an action, "IP was blocked on such date by such user," and you can save these things and that gives comments that can be viewed by other people working on this Case or other stakeholders, or just to memorialize data that you think needs to be memorialized, for after-action review. It may be that you need to go back for compliance reasons and review these things and anything you can put in here is going to add value to review later on. **Evidence Locker** is a way that you can attach files to Cases. These are very useful functions.

So you can add a CSV of information. It could be a data table that you've exported. It could be information from another event that is similar that has a bearing on this Case.

16:00-17:03
It could be a PDF that you've exported, some report that you've created about a certain type of activity that is involved in this Case to give context for other consumers of this information about what is happening in this Case and what might be happening in other parts of the network that relate to it. Also .eml files can be attached. So if there is correspondence going back and forth between different stakeholders, different teams, potentially external reviewers of this data, you can attach these email files and that can serve as very useful context for any further analysis that needs to go on or for the current analysis that is going to allow you to make an informed decision about how to take a response action against the Case. Thanks for attending this demo. There will be more demos to come in future lessons.

The lesson, *Demonstration 2 – Example Investigation of a Case with Tools in the Stellar Cyber UI* is complete. You should now be familiar with some of the tools that are available to you when conducting investigations. Thank you!

L02 - ALERTS IN THE UI

L02A-PRES: Alerts in the UI (07m:00s)

00:00-01:00
The next lesson in the *Introduction to Features of the Stellar Cyber User Interface* module is *Alerts in the User Interface*.

Alerts in the User Interface. In this lesson, we'll take a look at what Stellar Cyber Alerts are, how they work, the Alert views we have access to in the user interface, how to navigate these Alerts effectively, and how to create custom Cases from individual Alerts.

In Stellar Cyber, Alerts are crucial components that help security teams detect and respond to potential threats. Alerts in Stellar Cyber are generated based on various parameters and data analysis. The system employs machine learning and analytics to compute Alert scores, which are influenced by factors like fidelity or confidence in the analysis; severity, or the importance of the event; threat intel, the reputation of IP addresses and URLs for example; and data period, the duration of the observed event.

01:00-02:00
Alerts in Stellar Cyber are triggered by machine-learning techniques and algorithms that examine raw event data for evidence of security breaches. Insignificant events are filtered out and anomalies are identified and generate Alerts. Stellar Cyber also leverages Alerts from third-party services through native Alert integration. Stellar Cyber's interface allows users to filter and view Alerts based on multiple criteria such as tags, source, XDR kill chain stage, Alert score, and status. The Alerts table provides a comprehensive list of Alerts, which can be sorted and filtered to help analysts focus on the

most crucial threats. Additionally, Stellar Cyber uses machine learning not only to generate Alerts, but also to classify and correlate them, improving the efficiency of security operation centers.

02:00-3:00
This AI-driven approach helps in detecting events within the network by correlating and combining Alerts, ultimately aiding in the transformation of correlated events into Cases. These features underline the sophisticated capabilities of Stellar Cyber in leveraging advanced technology to assist security teams in proactively managing and responding to cybersecurity threats.

So how should we think about triaging and investigating these Alerts? We spoke in the Cases module about how it's an effective technique to take on the Cases first because Cases have lots of individual Alerts within them and by triaging and investigating all of these together as a comprehensive event, you knock out a lot of the Alerts that you will have left once you get through your Cases. So you have some things to decide when it comes to creating a workflow, and we will discuss these factors in the workflow modules. But, for example, you need to decide if you want to triage all of the Alerts, or if you want to develop a method of processing the most important Alerts and leave some unexamined or minimized.

03:00-04:00
Factors for this could include, do you have the workforce resources to cover all of your Alerts every day. Also, what is your organization's risk tolerance for certain types of activities? You may not want to process all Alerts, but you may still want to see them all. These pages we'll see are a critical place to do tuning on your system as well. You're able to examine what types of Alerts are noisy, which ones you want to consistently review, and if there are misconfigurations on the platforms that feed data into your Stellar Cyber instance. For example, Stellar Cyber, because it aggregates many types of logs, allows you to see things in a different perspective than in the individual security platforms that you're ingesting on their own. So it may show you things that other people don't see. You may see Alerts that trigger that tell you that there is possibly misconfigurations in some of the other platforms. And so Stellar Cyber has a value add of being able to see things in a slightly different way by aggregating these things and showing you the data in ways that you might not see it if you're looking at it in a single platform. One of the views we'll see in this presentation is the place to do your threat hunting as well. So you should think about Alerts as a place to do your threat hunting. These Alerts, especially the grouped view on the Alerts page, is a great way to see what's happening on your network in an efficient way. On this page you can see the type and number of each set of Alerts and determine if there is any tuning that needs to happen in Stellar Cyber or tuning in any of the platforms that provide data to your Stellar Cyber instance.

04:00-05:00
Alerts--How Does It Work? Stellar Cyber presents Alerts in a way that is designed to help analysts efficiently investigate and respond to threats. The system typically aggregates and correlates data from various sources to detect anomalies or suspicious activities, then generates Alerts to notify analysts of potential threats. Here's how Stellar Cyber presents these Alerts to facilitate threat investigation.

Aggregated and Correlated Alerts Stellar Cyber combines data and generates Alerts from multiple sources and uses Graph Machine Learning algorithms to correlate Alerts into Cases, reducing the number of false positives and highlighting more relevant threats. This aggregation helps analysts focus on significant Alerts without being overwhelmed by noise.

Prioritization Alerts are prioritized based on their severity, impact, and the likelihood of being a legitimate threat. This prioritization helps analysts focus on the most critical issues first.

05:00-06:00
Contextual Information Each Alert is enriched with contextual information to provide analysts with a better understanding a threat. This can include details about the affected systems, user accounts, network traffic, and any related events that might help assessing the situation.

Interactive Dashboards. Stellar Cyber offers interactive dashboards that allow analysts to visualize the data and understand the scope and scale of an Alert. These dashboards can be customized to highlight key information and trends that are relevant to the organization.

Investigations Workflows Stellar Cyber provides structured investigation workflows that guide analysts through the process of examining and responding to Alerts. This can include automated steps, such as gathering additional data, as well as manual steps, such as isolating affected systems.

Integration with Other Tools Stellar Cyber can integrate with other security tools and systems, allowing analysts to leverage additional data and response capabilities. This integration can help in automating responses and in providing a comprehensive view of the Alerts in your security landscape.

06:00-07:00
Threat Intelligence The system uses threat intelligence to enhance the Alerts with information about the known threats, vulnerabilities, and attack patterns. This helps analysts quickly identify the nature of the threat and the potential tactics, techniques, and procedures being used.

Alert Management Analysts can manage Alerts, mark them as resolved or escalated, and document their findings and actions. This helps in tracking the response process and in improving the overall security posture over time. By presenting Alerts in this manner, Stellar Cyber aims to reduce the time and effort required for analysts to identify and respond to security threats, ultimately enhancing the organization's ability to detect and mitigate cyber risks.

L03 - ALERT MANAGEMENT

L03A-PRES: Alerts Views (08m:26s)

00:00-01:00
The next lesson in the *Introduction to the Stellar Cyber User Interface* module is *Alerts Views*. *Alerts Views*: In this lesson we'll take a look at the Alerts views that are available to you in the Stellar Cyber

User Interface. Alright. In this lesson we're going to be looking at the Alerts views that are available to you in the Stellar UI to navigate and triage and investigate the Alerts that are, in theory, left over after you've done the investigation on the Cases and eliminated all of the Alerts that are associated with one another and closed them out or escalated them or achieved whatever outcome you're looking for with a specific Alert that happens to be in the Case. And then we'll attack the Alerts one by one or we'll group them together and we'll talk about different ways that we can either group them together or analyze them individually, to accomplish all sorts of goals depending on your security needs.

01:00-02:00
If we start from the UI, first we have Alert metrics. So that's going to show you the number of critical Alerts that you have to triage and the total amount of Alerts that are generated in your system on any given time period that you set. And next we will click directly into the Alerts view page and that will show us all of the Alerts in our system that are grouped by Alert type.

So in this view, you have all of the Alerts grouped by type. And this allows you to do all sorts of analysis. You might go in and see if there are any commonalities in the Alert types that you might want to group into a custom Case, and it gives you an idea of what sorts of Alerts are generated in the time period that you're viewing. And these Alerts are all filterable, and you can configure this field in any way you want. On the left, highlighted here, we have a set of fields that you might want to add. This is a small amount of fields.

02:00-03:00
There are ways that you can put any field that you want in this table view and you can arrange them, configure them, filter them, however you want to do that, and we'll share that during the demo.

The second view that's really important is under the **Investigate** tab, you have a threat hunting option. And what's different between this view and the Alerts view is that this is all of the Alerts for your system. So it allows you to sort and configure in a different way so that you see all of them in one view versus having them grouped by Alert type. And there's lots of reasons to use this particular view, which we'll explore in the demo, but it typically facilitates comprehensive manual analysis of the views that you want to investigate. So the Alerts that you have, you can sort them, group them, add fields, change the fields, each type of Alert is probably going to take a different set of values in the fields that you choose, and we'll talk about that during the demo as well.

03:00-04:00
Another thing that you can do, if you find something that does have commonality, you can create your own custom Cases. And this can be done so that you can investigate it yourself all at once or it might be that you want to escalate this and put it in the queue for your SOC analyst. So here we have the same process user in this “System DLL Overwrite” Alert four different times. We may want to create a Case out of this and knock these all out in one investigation. Here the process user MaxEzekiel is responsible for four DLL overwrite Alerts; and we'll talk about what that is in just a minute. We need to investigate this activity as a comprehensive event rather than one at a time. It just makes more sense and saves you time and energy. So any type of activity that has

commonalities among the fields can be combined into a Case for investigation by the creator of the Case. Or it can be added to the queue for SOC analysts to investigate. We'll see more of this type of activity in the demo, but having an analyst look through the Alerts manually to correlate the events that don't automatically create a Case is a very viable strategy to build workflow around.

04:00-05:00
The commonality may be a single type of Alert like you see on the **Alerts View** page. Grouping Alerts, even though they may not have any other common entities, is a valid strategy to find out what is causing a specific type of Alert in your environment. This strategy definitely promotes strong security hygiene and health.

So here we can see that a new Case has been created and the fields that you see in the Case versus the Alert view may not be the same and we'll talk about how to customize the data table so that you can have all of the fields for a specific type of Alert. Because each type of Alert is going to require analysis on the relevant fields and the relevant fields may not be the same in all Cases. So here, we have your MITRE Enrichment for the Alert, which is a really super powerful tool to tell you exactly what's happening. So not all analysts know all attack modalities and having this MITRE ATT&CK framework data here lets you know what this is.

05:00-06:00
In this Case, this person is trying to overwrite a DLL. So a system DLL overwrite refers to a scenario where an attacker replaces a legitimate system dynamic link library, or DLL, file with a malicious one. Since DLL files are essential for running various applications and processes in an operating system, overriding them can lead to serious security vulnerabilities. So when a system DLL is overwritten, the malicious DLL is loaded by applications or the operating system itself, thinking it's the legitimate library. So this can allow attackers to execute arbitrary code in the context of the application or the operating system, leading to an unauthorized action. Like in this Case, privilege escalation, but also potentially data theft, system compromise, or spreading malware. And doing the investigation in this MITRE ATT&CK framework lets you know exactly what this is. Obviously if we didn't know what a DLL was, we wouldn't recognize that this is a very dangerous operation that this person is trying to carry out.

06:00-07:00
We know that this one user is trying to overwrite DLLs and probably conduct some sort of malicious activity on your system. So having the knowledge and being able to investigate these four at a time versus one at a time is a very valuable tool for efficiency in the SOC.

So we talked a little bit about filtering a minute ago, but you can filter this content. And in the Alerts view on the side, you can see that these filters are by default, and you can add extra filters if you want to. So you have all of these options: tags, Alert options, that means if there is an Alert or if there is even no Alerts for that type, the source of the Alert, the status, the kill chain stage, asset tags that have been added, assignees for who has been assigned this task, event tags, fidelities, queries, the sensor it came from, and the severity of the Alert. So all of this gives you really valuable information in order to start your investigation or conduct triage on a large number of Alerts.

07:00-08:00
Let's take a moment to check your knowledge.

What is the difference between the Alerts view and the Investigate, Threat Hunting view?

1. The **Investigate > Threat Hunting** view is not customizable.
2. The **Investigate > Threat Hunting** view groups Alerts by type; the **Alerts** view displays all Alerts ungrouped.
3. The **Alerts** view groups Alerts by type; the **Investigate > Threat Hunting** view displays all Alerts ungrouped.
4. The **Alerts** view is the only view that displays all Alerts in the system.

The correct answer is C. The Alerts view groups Alerts by type; the Investigate Threat Hunting view displays all Alerts ungrouped.

True or False? Custom Cases can be created from Alerts that share commonalities.

08:00-08:26
The correct answer is True. If you find Alerts that have commonality, you can create your own custom Cases.

True or False? MITRE ATT&CK data is integrated into the Alerts view to provide additional information about the nature of the Alerts.

The correct answer is True. The Alerts view includes MITRE ATT&CK data, providing you with additional information about the nature of the Alerts.

The lesson, *Alerts Views*, is complete. You should now be familiar with the Alerts views that are available to you in the Stellar Cyber User Interface. Thank you!

L03B-DEMO: Interacting with Alerts (21m:26s)

00:00-01:00
In this lesson of the *Introduction to the Stellar Cyber User Interface*, we're going to be doing a demonstration of how you interact with Alerts in the Stellar Cyber user interface. The views we're going to take a look at today are the **Kill Chain** view, the **Alerts** page view, and the **Investigate | Threat Hunting** view. All of these views vary slightly, but they all allow you to get to certain elements of information that you need to conduct adequate investigation and triage of these Alerts. lThese views are not presented in order of importance. They are all equally important in the way that you conduct your workflow, and we will talk about workflows in the *Workflows* module of the *Introduction to the Stellar Cyber User Interface* module, but we will address best practices for each of these views in your daily operations.

Alright. So we'll get started triaging Alerts in the **Kill Chain** view. In this view we take a look at the stages in the attack **Kill Chain.** From left to right we go from less dangerous to most dangerous operations within the **Kill Chain** and we'll take a look at them real quick.

01:00-02:00
If you hover over these, you see the tactics that you'll be experiencing. So in Initial Attempts we look at where attackers are trying to access your network. Then Persistent Foothold, that's where attackers are trying to maintain access to your system in spite of defensive techniques. Moving on to Exploration. We're seeing attackers inspecting your network, learning about your environment before taking further action. In the Propagation phase, attackers are trying to obtain additional privileges and access and control additional systems on your network. And in what's considered to be the most dangerous stage, Exfiltration and Impact, attackers are trying to steal data, potentially dumping it in a way that avoids detection, and or vandalizing your systems and data. So within these stages of the **Kill Chain** we see the different tactics that they're using. So for each of the stages of the **Kill Chain,** you have different tactics involved in that stage of the attack.

02:00-03:00
Using this view to process Alerts assumes that you've gone through the Cases and eliminated as many related Alerts as you can beforehand, and then this number shows you the number of critical Alerts versus the total number of Alerts that have not been closed. So if you choose to integrate this into your workflow, you might go and start with Exfiltration and Impact first, because it is theoretically the most dangerous phase of the attack chain. And we'll take a look at these and just auto size all these columns so that we can see the results better. And once again, I'll remind you that the columns are editable. You can pick either through this menu right here, which is a limited number, or if you went to view the Alert itself, you could go into the metadata associated with it and add a field so that you can see the fields that you want. And each type of Alert is going to require different fields for you. So you can set that up for each type. If we choose to do the analysis on the Alert itself, like say internal ransomware, that's a very dangerous Alert.

03:00-04:00
We want to see what's going on here. And here it tells us **Kill Chain** stage Exfiltration and Impact, impact is the tactic, data encrypted for impact, and it's telling us that malware that prevents you from assessing your system or files and demands ransom payment in order to regain access and internal traffic was observed. Check with the user. So here we have a limited set of fields that we're looking at here, and we'll auto size this again. So we can see that same description with the IPs and what it is, but maybe we need to add more fields in here so that we can do this analysis. So we might want to add the severity, and we might want to add the action that was taken. So event source Sandbox we're going to show this column because we want to know that this was in a sandbox and not in our environment. We'll also want to add the outcome from the maltrace, which is where we're seeing this from. Let's see if there's any other information that we might want to see in our view.

04:00-05:00

You might want to see the technique as well in here, and the event name, and the username. So now we have a better view of this in our fields. So that if we wanted to look at this in the data table view that we're seeing here, we have more information. The Alert score is a hundred, so it's clearly dangerous. It's telling us what the malicious activity is, a good description, then we have the source, which tells us that it's in a sandbox and not going through our network, and then the vendor's outcome here. We've got success so we know that it's not as dangerous as we thought it might. And here we have a number of visualizations that show you information about this particular event. In this case, these visualizations are not particularly useful, just because it's only one Alert, but if we had a group of Alerts, you'd see some valuable information here.

05:00-06:00
And if we click **Investigate** here, we see that this ransomware was deployed in the sandbox, and we see the same graphics. But if we were to click over on other malware, this is what a useful visualization would look like. Here you see the source host, as well as the IPs that need investigation. And so this is what it would look like if there were more than one Alert to be seen here. So now it's time to take action on this event now that we understand it. So to get to the actions, you go into the **info** field. And click on **Actions.** And first of all what we want to do is block this IP. So fortigate is our firewall. The IP it shows here for handy reference. So we'll block this incoming forever and click **Submit**. And also what we'd like to do is trigger an email to the user.

06:00-07:00
So we want to send this person an email to contact us and we would give them the information. We might reach out directly to them just to make sure that nothing else malicious has happened, and we might want to find out how this happened to their system in the first place. Now that we know that this has been contained, we want to go ahead and close this event out. So what we do is we just check the box, and then go to status, change the status to closed, and apply it. And if we go back to our homepage, we now see that there are only five of 16 Alerts total. So we've had some new Alerts show up. But we see that the closed ones do not show when you're clicking through this view. So the takeaway from this particular view is that maybe after you've cleaned out all of your Cases and all the associated Alerts with the Cases, you might start with the highest-interest **Kill Chain** stage to your organization and check that out.

07:00-08:00
See what's going on with regards to propagation on your network, who's doing exploration, who's seeking a persistent foothold, and then maybe go to Initial Attempts because a lot of the time your security vendor products are going to prevent these initial attempts, but it is also worth investigating to see if any of these attempts are successful so this is how you use the **Kill Chain** view You basically just pivot from the **Kill Chain** graphic itself into the Alerts triage the Alerts, respond to them, and then move on.

Alright. Now we are going to move on to the **Alerts** view. Clicking here, and here we see that we have a group of Alerts, and we'll go ahead and autosize these columns so we can see all our field names. Got a filter here. Let's unfilter all these. So let's start here with the ability to sort by column.

08:00-09:00
And once again, we have this limited set of columns that we can choose, but just like the other ones we can go into the Alerts themselves.

And if we want to add a column, we just add it here. So let's see. The **Process User** is probably important. Let's show this column. We'll put it here and then move it over here. So the same process applies to the **Alerts** page where you can sort and stack. You can move these columns right and left. You can build a table with the fields that you need for any specific type of Alert. Again, it's possible to select all of these systems and create a new Case to it. If we wanted to add this to a Case to investigate all of the system processes that are running, or if we have a person who has lots of events we could create a Case out of that and do that investigation that way.

09:00-10:00
The main thing to know about your **Alerts** view page is that the Alerts are grouped by Alert type. So you'll see that there are 1,389 events here. That's a lot. So it's important that you're aware of the events that are going on on your system to know what's important and what's not.

So another thing to know about the **Alerts** view page is that you have a lot of filters accessible to you. So you can add a custom query here. Anything you want to create you can use. To filter this by just a specific query. If you have a specific event type that you're looking for, any tags that you have made, custom ones or ones that can populate it out of the system. **Alert Options**: What this means is show Alert types without hits.

So where this is probably useful is if you show Alerts without hits and this is how you add Limit of events for your page.

10:00-11:00
So I set this to 400 because there's 358. We have a lot of events here that don't have any results. So you might think, "Well what's the point of this?" This is very useful for you in case you want to see what Alerts should be firing in your environment but might not be. So this is a very useful view to see what's going on in your system versus what's not going on in your system. So we will turn that off so we don't look at that. Then we can filter by this slide bar Alert score. We just want to see from 80 to 100 for example, and that gives us less results. Then we might want to filter by the source, either built in or custom Alerts. And you can filter by Status. Likewise you can filter by the **Kill Chain** and then the tactics within the **Kill Chain.** So if we want to just look at Exfiltration and Impact, we have essentially the same view that we had when we went directly from the **Kill Chain** into the events that were in each stage of the **Kill Chain**.

11:00-12:00
You can get granular here and choose the tactic within the **Kill Chain** to tailor your results.

Also, like the other views, the Case views you can add a new filter. So any of these options are able to be filtered on. So you can filter by severity, filter by threat intel, usernames, all of this is available to be used as a filter. So that's very important when you're dealing with large datasets. Or you might

want to give specific jobs to specific members of your team. And this way, they can filter and get only results that they need and they're not stepping on other people's toes,l duplicating efforts, very useful in this way.

So if we go back to the **Alerts** view page, we can see that we've got these Alert types grouped together. And the first question is probably, "How do I triage 1,389 events in a single Alert type?"

12:00-13:00
That's a little bit outside of the scope of this particular lesson. We'll talk about that in the *Workflows* module. But for purposes of looking at how Alerts look in the UI which is our topic today, we'll pick something manageable. One thing that's good to know about the **Alerts** view page is that you do not have these things sorted by criticality. So it's sorted by number, generally. So it doesn't give you the Alert score. So you might look at something and say, "Well let's see internal ransomware. That's kind of a big deal. Maybe it should be up top, but that's not the way this view looks." The view is intended to give you the number from top to bottom and you use your own discretion to decide which ones need to be looked at more closely. When we get to the **Investigate | Threat Hunting** part of this lesson, we'll see how you can actually sort by Alert score and see all of your Alerts not just grouped by Alert type.

13:00-14:00
So we'll start with credential dumping. Everyone should know what credential dumping is. That's where they steal your credentials and use them to do nefarious acts on your network. So we'll take a look at these and see what it looks like. We first of all have a chart with the time stamps. We have high fidelity all of these. There doesn't seem to be any location information, or we'd see stuff on this map. So we can tell that this Alert is coming from CylancePROTECT, and it looks like there is no host name, host IP, filename, or file path in this particular set of data. So what we want to do, is go into this metadata here and find the things that we are interested in, because we've got four Alerts here to triage, we might have 1,300 in another Case. So we want to get our data table, giving us information that we want. So we'll look through these real quick. The blocked action is something that we definitely want to know.

14:00-15:00
We definitely want to see whether this is blocked or not. We also want to see the Process User, the system that means that they have the highest privilege as possible. So this tells us we may want to go in and look for other instances where someone's gained access to a system process user. We want to add the Tactic. And the Tactic ID, the Tenant that we're using, and the Technique. And we'll get the Event TTPs as well. So now that we've got our table fields like we want them, we might want to change this around to have the action that was taken on here first, so that we might want to go look for things that weren't blocked. So really quickly, this will be addressed in the module on how to create queries, but an interesting thing about the UI that's very helpful, is if you want to filter for, or filter out, you can do that with filter-lout blocked–

15:00-16:00

so we know these are all blocked with the exception of the none. So we might want to take a look at the none. But if we want to create a query, we can copy this, and it will give us everything that is blocked. So in this particular case, we might want to say, we might want to see everything that was not blocked. So we're going to exclude this using the UI. So we're going to say filter out all the blocked events. And so now we see we have our one result. But if we look up here, what's happened is we've added a filter. The system has already added you a filter without you having to do it. So we'll take a look at this query as well.

16:00-17:00
You can copy this as a query. And the syntax will look like this: NOT blocked or not keyword blocked. And that's how you create queries. You could run it like this, and we'll get the same results here. But this is how you basically create your queries from the data that you actually see in the UI. You can create these show column, filter, just for none, filter out none, copy this value to the clipboard, or copy as a query. So here we'll see what the query looks like up here again. So now we see it phrased in the positive. So if we query this, we'll get the same result here. And that's how it works. Here now we've created a table that contains the information that we want. We might want to exclude this field and this one since there's no values in it.

17:00-18:00
And now we've got something that we can use to do these investigations, and maybe we start with none. And see why that's happening. So that's how you interact with data tables in this way.

Also, I would add that you can search for a specific value here and filter out all these things just by a keyword. And like the other Alerts, you can go to Actions here. You can see the key fields, any Cases that it's involved in. So theoretically this would have been already triaged. Your details are all here. This Interflow data is what you create queries out of. You may have noticed that there were two values here. They were both derived from this data. So **Event Name** and **Event Name Keyword** that is covering both permutations of how the data might be indexed, so that's just making sure that you don't miss anything.

18:00-19:00
And then you have your Actions. You can take an action on any of the values in these fields and eliminate the threat before it becomes a real problem for you. Also I would remind you that this information here is very important. And these are hyperlinks to take you out to the MITRE page. And you may not know what LSASS Read means, right? You may not know that LSASS stands for local security authority subsystem service. It's a crucial component of the Microsoft Windows operating system that is responsible for enforcing the security policy on the system. So it basically allows you to reset passwords. It could be used to dump credentials. So given its critical role in managing security operations, LSASS is a target for various types of attack including, but not limited to, attempts to extract sensitive information like passwords from its memory, or to exploit vulnerabilities for gaining unauthorized access or privileges to the system.

19:00-20:00

So very dangerous. If you don't know what this is, it's very important that you look it up. So not every analyst can know every single thing in the whole world. There's no shame in looking things up. And these are your valuable resources within the UI that give you that information. And tell you that if I don't know what LSASS is, I may not evaluate this as a danger. But if I do, I know that this is a very dangerous thing and it should be taken very seriously.

So the last few we're going to look at is the **Investigate | Threat Hunting** view. And there's a lot to digest here. You've got the same filters. You've got a summary shown in visualization form of a lot of different kinds of data. You also have an Interflow search, which is what we looked at a minute ago. Correlation search is where you can build correlation queries. And then **Threat Hunting Library** is a set of prebuilt query results that you can see different types of information. There's lots of them there.

20:00-21:00
So because processing Alerts in the UI is the focus of this lesson, we'll look at the queries in the *Query* module later on. But the main point of this view is that this gives you all Alerts in your system. And so if you develop a workflow that requires you to triage every result that you get, so every Alert, every event that shows up in the UI, you might use this view. So **Investigate | Threat Hunting**. This view shows you all of them. So if you develop a workflow that says, "I am going to sort this by Alert score, and I'm going to either triage every Alert that's between 80 and 100, or if you have the workforce, do every single Alert that pops up." So you can filter here to get the ones that you want. And essentially all of the Alerts function in the exact same way as they work on the Alerts page. So we don't have to spend a ton of time.

21:00-22:00
But the key takeaway here being that you can sort by Alert score, you can filter out Alerts, and this gets you everything without the grouping of Alerts, which can get in the way if you're doing some sort of special analysis. So this is a really useful view to really drill down into the Alerts and create custom queries and to see what's going on in your network.

L04 - THREAT HUNTING

L04B-DEMO: Threat Hunting (19m:14s)

00:00-01:00
Welcome to the *Threat Hunting* lesson in the *Introduction to the Stellar Cyber User Interface* training module. Today we're going to be looking at how the threat hunting features look and how they work in the user interface. We'll start by looking at filters and show how filters are integral to getting to the data that you want to see and not the noise that you don't. We'll look at the query bar and take a look at how the user interface integrates this query-building capability into the user interface. We'll take a look at the Interflow search and what that is, and then we'll look at the threat hunting library, which is a really useful interface to have prebuilt queries. And then we'll take a look at the Interflow dictionary and what that is. So let's get going.

Alright. We talked a little bit about the **Threat Hunting** option under the **Investigate** tab. So we'll navigate there.

01:00-02:00
First let's check out the filters options. So by default you have **Alert Score**. So an example of the use of this might be is once you've got your Cases cleared out and you want to just examine the most high-priority Alerts, you can filter by score, and then examine each one of these one by one. That's one potential use case for the **Alert Score**. And then we have **Asset Tags**. So if you have given tags to your assets, like say you have domain controllers and you want to give them a tag of critical systems, that's one way to filter out all of the results that are in the **Investigate | Threat Hunting** tab.

So next you have **Asset Tags** and there are not any assigned in this test demo system. But if you wanted to give your critical systems a tag of "critical systems" like say you wanted to label all domain controllers or label all the servers that contain really sensitive customer data, you can give them a tag and then filter by that,

02:00-3:00
so you can get only the result for your critical systems. That's one use case of this. **Status** is pretty obvious If you want to only view the new Alerts or if you want to go back and review all of the closed Alerts that have shown up in the **Threat Hunting** view. You can click closed and then go make sure that no mistakes were made and no threats were closed out. And then as in the other pages you have an **Add new filter** option. So If you just click in the box here, you could add any of these. And say you only wanted to view a certain command string. So you can use that and then you have a command string and you could filter on that, or you can filter by users, or any of these. You've got a lot of fields here that you can use to filter by. So it just depends on what you're interested in. But you can set a filter so that you can filter just the results you want to see. Maybe Destination IP you want to see if you can find a destination IP that's getting hit with ransomware and look at all of the different activity on that particular account. You can filter that out and only see what want to see,

03:00-04:00
but the sky is the limit here as far as what you want to filter on. Pretty much any field that has data in it you can filter on it.

Another cool option here is if you want to convert any of these filters into a query, we talked about the search bar. You can copy this as a query and then just paste it in as your parameters, and then click **Enter.** And once again we've got this as a query so that you only have the results that you want to see.

Similarly with **Queries**, you can create your own queries and filter on those. So we'll view more here There's not a lot here in this demo lab, but you can create your own query if you want to. So we want to create a new query that says, "I only want to see this specific activity that I have seen in my results," and you can add any condition that you want.

04:00-05:00

This field contains this value or is exactly contains field exists field does not exist, starts / ends with, is in lookup, or is not in lookup and all of these allow you to get tailored results for any type of data that you specifically want to see.

So here if we wanted to for example create a query where we go look for all commands that delete shadow copies, so we want to find if anybody's deleting shadow copies, we want to see all of that activity. So we'll search for it. No results here, but you can copy this as a query and save it if you'd like to. And then you can have access to that anytime you want to see it. So this creates it for you and saves you the trouble of having to look at the query syntax. It's very easy. So we don't have any results here, but if you had results, this would be the way to search for it. And you can save that query and use it later.

05:00-06:00
And another way that I will show you to create queries, let's get rid of these arguments here, is if you want to, just like in the other views, you can copy this as a query. If you want to have Alert tags custom as your filter, or if you want to see all PLC checksum changes, you can copy this as a query, go back to your search bar, and your query is built for you. So now you only have just the PLC checksum changes. And you can do your analysis this way. It's a really effective way to filter your results by just using this **Query** button and it's all built into the UI for you. You don't have to search what the syntax is on your own.

Next, we have the visualizations that appear at the top of the page. You have your **Top Sensors** and you can see which ones they are just by hovering over the graphic. Your **Top Log Sources**--this tells you what sort of traffic you're getting in your system that triggering Alerts.

06:00-07:00
You have a timeline of how much activity is happening at any given time. And this is particularly useful if you filter on a certain activity type. So say I want to see when there's checksum changes across my network, I can just filter that and then this graphic will represent that. And then you have your **Top Log Source Classes**, and these are types of activity It could be audit rules, Interflow traffic, could be firewall data, IDS files, powershell activity, any of these you can filter on. You can see what types of activity is occurring on your network at a glance.

Alright, let's talk a little bit about **Interflow Search**. You can see that Interflow search is here. This is the tab that we're under on the **Threat Hunting** page. And what this means is basically you can create these searches like we saw just a minute ago, and there are a couple of important things to know about the **Interflow Search**.

07:00-08:00
If we want to search for this particular anomaly ID, we can just click this **Copy as query**. And one of the things that we'll notice in the query is that the field name, “anomaly ID” is not exactly as it appears here. You see that there's no underscore here. In this query, we have an underscore. So where that comes from is if we go look in the information view of this Alert, we'll see that there is a set of Interflow fields and values. If we look for anomaly ID here, we see that it shows up here, but it

doesn't show up here. So if we search for “anomaly,” we find that the field name in the Lucene syntax version is anomaly underscore ID And this is how you look up the field to create queries.

08:00-09:00
Another important thing and you'll see this in some of the results, is that you may have a more complex query. So let's see. So the parent process name here, we'll copy this as a query and see what the query looks like. And here we can see there's two fields here. One of them is parent underscore proc underscore name and then systemMD or parent proc_name.keyword, and these populate because there is a slight difference in the way the fields and values are indexed in the database. So in order to get all of your results, you want to use this format that if a field has a dot keyword, you'll use that as well and that gets you all of your results. And we'll go deeper into the query syntax in the *Query* module. So don't worry, this is not a comprehensive view of the way you construct queries.

09:00-10:00
That will be covered in a different module. But I wanted to explain that this is why your query might be a little bit more complex than you would expect it to be if you've just found the field and paired it with the value that you're looking for.

Now we'll switch tabs and take a look at the **Threat Hunting Library**. The **Threat Hunting Library** populates results through a combination of pre-built playbooks, automated execution, and integration with the platform. It leverages the power of automation, Normalization, and Enrichment of data. Here's how it works. The library contains over forty automated threat-hunting playbooks that cover a wide range of threat vectors and attack surfaces, including but not limited to, Windows login failures, DNS analysis, Office 365, things like that. These playbooks are designed to automatically search for behaviors of interest ensuring comprehensive coverage of potential threats without the need for constant manual oversight. So we'll take a look at these.

10:00-11:00
We don't have time to go through all of these because they're very comprehensive, but we'll take a look at Login Failures. So we've got login failures over time, how the login failures are happening. Top users and here we say Aella, is the top login failing user, and then this IP. So we might want to take a look at these results. So these are Windows. So we may want to drill down and see what exactly is happening with this Aella user. So we'll just create a query here. Or we could filter for this user and see what's going on. So we'll just filter here instead of using a query. So now we can examine these failures. Remember to remove your filters or you'll have zero results.

So we'll look at **Linux Privilege Escalations** and see what's going on there. We have privilege escalations by top users and Zhou is the top user, so we want to see what's going on with this particular user.

11:00-12:00
And in this way, you can drill down and see the exact data that you're looking for So here we see that Zhou is showing up in the primary user. So if we just filter for this, we can see all these Alerts. We might want to group them into a Case If they're not and we can check and see if they're in a Case

already. Alright These are not. So we can examine these. It looks like this is red team activity so it might not be as dangerous as we think. But this is the way that we would examine this type of behavior. And remember to clear your filters.

Alright. We've got one last thing to take a look at in this lesson and that is the Interflow dictionary. So under the **Threat Hunting** page, we see here that we have **Open Interflow Dictionary**. So we're going to take a look at what that is. So the Interflow dictionary in Stellar Cyber is a tool designed to aid in threat hunting by providing a comprehensive reference to the metadata fields used in Interflow records. And all of the records that are in the platform are converted into Interflow records, which means that they are parsed, and they are Normalized, and they are put in this field,

12:00-13:00
and we saw it when we were looking at. Let's take a look at one of the Alerts here. So we saw this here under JSON. We have all the all the data that we can see. So we have key value pairs here engine device class is Windows. This is the key, and this is the value. So when we do have data it's pretty easy to sort out which field, which data goes into. But if you don't have that information, or if you just want to be aware of all of them, we can go into the Interflow dictionary which is under the **Threat Hunting** page. And here you have all of the fields that are possible. And this helps you understand the various fields that can be queried or filtered within the Stellar Cyber platform, and it makes it easier to construct precise queries and effectively analyze security data.

13:00-14:00
It's particularly useful for identifying the context and significance of different data points in the Interflow format. It really enhances the efficiency and accuracy of threat detection and investigation process to have all of these and you also have information about these. So this one is destination IP reputation. It's the reputation of the destination IP by threat intelligence. So it gives you the possible values, and it gives you the value description so you know what is expected in the input when you're creating these Interflow queries. It helps you create precise queries that get you the data that you're looking for.

And so over here we have a visualization of all the different types of anomalous failures. So we can change this to be 25 and you'll see there's a lot more once we get 25 in here. And so here we see that the **Accumulated Anomalous Failures** field, it indicates the score within the time weighted rolling window model that reflects the degree of abnormal activities. This field is particularly used in the context of machine learning to assess and quantify the level of anomaly detected on observed behaviors.

14:00-15:00
So here, here we have 22 and that represents the value of the field in the Interflow. And there were 314 events that represent this type of Alert. So the number three fourteen is the count of how many Alerts match that. So all of this helps you to create these queries and we can create these queries. If we closed the dictionary, we would use that in this field to create a query that got us the data that we wanted to see.

So creating Interflow search queries in Stellar Cyber--we spoke about these queries earlier and we'll explore them in more depth in the *Query* module--but it involves utilizing the search syntax to pinpoint specific records or events within the data that's collected in order to analyze it. So it helps to understand the data structure. So Interflow organizes data into records that are essentially JSON objects, and that's what we looked at when we looked at these other Alerts.

15:00-16:00
We have it here in the **Information** tab. This is the key value pairs that we have data for. And these records contain various fields or keys that represent different aspects of the data such as IP addresses, event types, application protocols, etcetera. So you should familiarize yourself with the common keys and what data they contain to know what you can search for. And this will become natural as you use the tool more to know which fields contain which types of data, so which keys contain which values. So in using the search syntax, it's designed to filter and find specific patterns within the Interflow data. So a basic search query might look something like...so here we might look for the source IP host. So if we search for this, we'll take these quotes out of here.

16:00-17:00
So here we have 14 results with that IP. And you know if this is not in a Case, we'll check and see if it's in a Case. Here it is in the Case now. So this looks like it might be all related to internal red (?) somewhere. Let's see. Yes. All of these are already grouped in a Case. So in our ideal scenario, which is we've cleared the Cases out first, we would have seen this and taken action on it earlier. So this syntax is flexible and allows for more complex queries that can include logical operators like and, or, not, and parentheses for grouping conditions so we'll see that in the *Query* module.

Another thing that you should know is that there is an **Indices** option here. These queries can sometimes be very resource intensive. And so by selecting a certain type of event-- like we want to see only Linux events--so here we don't have any in this lab. Let's see if we've got any malware sandbox events. So we have four.

17:00-18:00
So these are malware sandbox events and that might be a good way to do your threat hunting, is if you're looking for a specific type of activity you might set up a flow where you have someone go in and look for the malware activities that happened in the sandbox, and then maybe pivot out of that to malware type events that didn't get sent to the sandbox. So we'll just go back to Alerts here. So depending on what data you're looking for you might need to select a specific index to search within. That just tightens up your search query and makes it more efficient. So it organizes data into various indices like DP monitoring, Linux events, malware sandbox, like we saw. You can choose one multiple or all the indices to search through. And just remember that searching through multiple indices can be very resource intensive. So sometimes you should restrict your time intervals if it's very resource intensive to reduce the performance impact on the system.

18:00-19:00
With regard to the queries, you can further refine your searches by combining multiple criteria using operators for more precise results. So we might look for a specific IP and add another argument to

this query. So here we would be searching for exfiltration over alternative protocol. So what we do is just give this an "and," and group these two arguments like we talked about before. There's keyword and non-keyword arguments. So we'll group these, and then search and it comes back with our exact results. So that's how the search query works. And like I said we'll cover this in deeper depth in the *Queries* module. But that's effectively how it works. So now that we understand the Interflow dictionary, we can move on to the next lesson.

L05 - USER BEHAVIOR AND ASSET ANALYTICS

L05A-PRES: User Behavior and Asset Analytics (20m:30s)

00:00-01:00

The next lesson in the *Introduction to the Stellar Cyber User Interface* module is *User Behavior and Asset Analytics in the Stellar Cyber UI.*

User Behavior and Asset Analytics in the Stellar Cyber UI. In this lesson, you'll learn about the user behavior analytics functions within the Stellar Cyber user interface as well as the asset analytics.

Today's agenda, we're going to talk about the user behavior analytics functions within the Stellar Cyber interface. We'll also talk about the asset analytics, as well as the benefits and key components of both.

Stellar Cyber employs user analytics as part of its broader security operations and analytics platform. User analytics in Stellar Cyber typically involves the collection, analysis, and interpretation of data generated by users' interactions with networked systems and applications. This data includes login attempts, access patterns, data access and transfer, application usage, and more.

01:00-02:00

The primary goals of user analytics in such a context are to:

- **Detect anomalies** by establishing what normal behavior is like for each user, the system can detect deviations that may indicate a security threat, such as an insider threat or a compromised user account.
- **[Enhance Security Posture]** It enhances the security posture. Analyzing user behavior helps identify risky behaviors and can guide policy adjustments to improve overall security posture. For example, if a pattern of risky behavior is detected among certain users or groups, specific training or policy adjustments can be implemented.
- **Automated response**. In some cases user analytics can trigger automated security responses. For example, if a user's behavior suddenly matches known patterns of malicious activity, the system could automatically limit access permissions or alert a security analyst for further investigation.

02:00-03:00

- **[Improve Incident Response]** It improves incident response. By having detailed insights into user behavior before, during, and after a security incident, analysts can more quickly and accurately identify the root cause of the issue, assess the impact, and take appropriate remedial action.
- **Compliance**. User analytics can also help organizations comply with various regulatory requirements by providing detailed logs and reports of user activities. In the event of a security breach, these analytics can be crucial for forensic analysis and legal proceedings.

Stellar Cyber integrates these analytical capabilities into its wider security platform, which also includes advanced analytics, artificial intelligence, and machine-learning technologies to enhance detection and response capabilities. This integrated approach allows for a more comprehensive security posture that can adapt to the evolving threat landscape.

03:00-04:00

Stellar Cyber leverages advanced analytics and machine learning. User analytics play a crucial role in incident investigation through several key mechanisms. Here's how they contribute to effective incident investigation.

[User Behavior Analytics]

- **Behavioral baselining and anomaly detection**. Stellar Cyber's platform can establish a behavioral baseline for each user, which represents their normal activity patterns. By continuously monitoring and analyzing user behavior, the system can detect anomalies or deviations from these baselines. Such anomalies might indicate a security incident such as unauthorized access or data exfiltration attempts. Detecting these deviations early is crucial for prompt incident response.
- **Contextual insights**. User analytics provide detailed context around security Alerts and incidents. For instance, if an Alert is triggered by an unusual file access or transfer, user analytics can help investigators understand whether this activity is part of the user's normal job functions or if it's an actual threat. This context is essential for assessing the severity and potential impact of an incident.

04:00-05:00

- **Forensic Analysis**. During an incident investigation, analysts require detailed logs and activity records to piece together what happened. User analytics in Stellar Cyber can offer granular insights into user actions leading up to, during, and after an incident. This information can help identify the methods used by the attacker, the scope of the breach, and the data or systems compromised.

- **[Identifying Attack Paths]** It also helps **identify attack paths**. By analyzing user behavior, Stellar Cyber can help investigators identify how an attacker might have moved laterally across the network, for example. Understanding the attack path is crucial for remediating vulnerabilities, tightening security controls, and preventing similar incidents in the future.

- **[Improving Incident Response Processes]** It also helps **improve incident response processes**. Over time, the data collected through user analytics can inform improvements in an organization's incident response strategies.

05:00-06:00

By analyzing trends in user behavior and incident outcomes, organizations can identify areas for improvement in their security policies, training programs, and technical controls. So by integrating user analytics into the incident investigation process, Stellar Cyber enables organizations to respond to incidents more effectively, minimizing the impact of breaches and enhancing overall security posture.

The key components of user analytics within the Stellar Cyber platform reflects the comprehensive approach to cybersecurity. It includes:

- **User behavior analytics, or UBA**. This component analyzes the behavior of users to establish normal activity patterns and identify deviations that could indicate security threats such as compromised credentials or insider threats. UBA leverages machine learning and statistical analysis to detect anomalies effectively.

06:00-07:00

- **Risk Scoring**. Users are assigned risk scores based on their behavior and activities. These scores help prioritize Alerts and focus on users who pose the highest risk. It enables more efficient incident response and investigation.

- **Anomaly Detection**. Leveraging these advanced analytics, this component automatically identifies unusual user activities that deviate from established patterns. Anomalies could include abnormal access to sensitive data, unusual login times, or excessive data downloads.

- **Insider Threat Detection**. By analyzing user behavior, Stellar Cyber can detect potential insider threats. This includes identifying users who may be intentionally or unintentionally exposing the organization to risk, such as by leaking sensitive information.

- **Account Compromise Identification**. This involves detecting signs that a user's account may have been compromised, such as simultaneous logins from geographically distant locations, unusual after-hours activity, or uncharacteristic access to sensitive resources.

07:00-08:00

- **Forensic Analysis**. User analytics in Stellar Cyber provides detailed logs and activity trails that are essential for forensic investigations following a security incident. This allows security teams to trace actions back to specific users, understand the scope of an incident, and determine the cause.
- **Contextualization and Visualization**. Providing a visual representation of user activities and behaviors helps security teams quickly understand complex datasets. Dashboards, heat maps, and timelines are examples of tools used to contextualize user data for easier analysis.
- **Alerting and Reporting**. Customizable Alerts notify security teams about critical user-related incidents in real time, while comprehensive reporting tools allow for the generation of detailed analyses and summaries for review by security teams or compliance auditors.

08:00-09:00

- **Integration with Other Security Tools**. User analytics in Stellar Cyber is designed to integrate seamlessly with other components of the security operations architecture within your organization, such as threat intelligence, asset analytics, and incident-response tools. This integration provides a holistic view of security events and enhances the platform's overall effectiveness.

These components work together within Stellar Cyber's platform to provide a robust framework for monitoring, analyzing, and responding to user-related security events. It enhances the organization's security posture and reduces the risk of data breaches and other cyber threats.

Assets analytics in Stellar Cyber refers to the processes and technologies used to collect, analyze, and manage data related to organization's digital and physical assets. These assets can include network devices, servers, workstations, applications, and even cloud resources.

09:00-10:00

Asset analytics or entity analytics, makes up part of Stellar Cyber's user and entity behavior analytics capabilities.

- **User and entity behavior analytics, or UEBA**, enhances cybersecurity by using advanced analytics to identify anomalous behavior or activities that deviate from established patterns in user and machine behavior. This capability enables organizations to detect insider threats, compromised accounts, and targeted cyberattacks that traditional security measures might miss. By continuously monitoring and analyzing data from various sources, UEBA helps in early detection of potential security breaches, facilitating rapid response and mitigation actions. The goal of asset analytics within Stellar Cyber is to enhance visibility, security, and management of these assets through various means.

10:00-11:00

- **Asset Discovery and Inventory**. Stellar Cyber automates the process of discovering and cataloging all assets within an organization's network. This includes both authorized and unauthorized devices. By maintaining a comprehensive inventory, the platform ensures that all assets are accounted for and monitored, reducing the risk of blind spots in the security posture.
- **Vulnerability Assessment**. Asset analytics involve continuously assessing the security posture of each asset by identifying vulnerabilities, misconfigurations, and noncompliance with security policies. Stellar Cyber can prioritize vulnerabilities based on their severity, and the criticality of the asset to the business, which helps organizations focus their remediation efforts where they are needed most.
- **Behavioral Monitoring**. Beyond just understanding what assets exist, Stellar Cyber monitors how these assets behave over time. This includes tracking normal usage patterns and detecting anomalies that could indicate a security threat, such as unexpected data transfers, changes in configuration, or unusual network traffic patterns.

11:00-12:00

- **Risk Management**. By combining data on vulnerabilities, asset criticality, and behavioral analytics, Stellar Cyber helps organizations assess and manage the risk associated with each asset. This allows for informed decision making about where to allocate resources for security improvements.
- **Compliance Monitoring**. Asset analytics also supports compliance efforts by ensuring that all assets adhere to relevant regulations and standards. This can include automating compliance checks, generating reports for auditors, and identifying non-compliant assets for remediation.
- **Security Posture Improvement**. Insights gained from asset analytics can inform strategic decisions about security posture improvements. For example, identifying commonly exploited vulnerabilities across assets can guide the implementation of more effective security controls or policy changes.

12:00-13:00

- **Incident Response and Forensics**. In the event of a security incident, having detailed analytics on affected assets can significantly enhance incident response and forensic investigations. Asset analytics provide critical information about the nature of each asset, its data, network behavior, and any changes made, helping to quickly understand the scope and impact of an incident. Stellar Cyber integrates asset analytics into its broader security operations platform to provide organizations with a comprehensive view of their asset landscape. This not only enhances security and risk management, but also supports operational efficiency and compliance efforts.

13:00-14:00

In Stellar Cyber, asset analytics significantly bolster incident investigation processes. Here are some of the ways they contribute to a more effective and efficient investigation.

- **Comprehensive Asset Visibility**. Asset analytics provide a detailed and up-to-date inventory of all assets within the network, including devices, applications, and systems. This comprehensive visibility ensures that investigators are aware of all potential targets and entry points in the event of a security incident, allowing for a more thorough investigation.
- **Identification of Affected Assets**. During an incident, quickly identifying which assets are affected is crucial. Asset analytics enable rapid identification of compromised or at-risk assets, facilitating swift containment and mitigation actions to prevent further damage or data loss.

14:00-15:00

- **Behavioral Baselines and Anomaly Detection**. Stellar Cyber's asset analytics include monitoring the normal behavior of assets and detecting anomalies. This capability allows investigators to spot suspicious activities or changes in asset behavior that may indicate a security breach, such as unusual data access patterns, network traffic anomalies, or unexpected changes in configurations.
- **Vulnerability and Risk Assessment**. By continuously assessing vulnerabilities and risks associated with each asset, asset analytics help investigators understand how an attacker might have exploited these vulnerabilities. This insight is critical for determining the attack vector and for prioritizing remediation efforts based on the severity and impact of vulnerabilities.
- **Contextual Information for Forensic Analysis**. Asset analytics provide rich contextual information about each asset, including historical data on configurations, connections, and modifications. This information is invaluable for forensic analysts, helping to reconstruct the sequence of events leading up to and following an incident.

15:00-16:00

- **Enhanced Incident Response**. The detailed insights from asset analytics allows for a more targeted and efficient incident response. Knowing exactly which assets are involved and how they are affected enables responders to quickly implement specific containment and remediation strategies.
- **Post-Incident Recovery and Hardening**. After an incident, asset analytics can guide the recovery process by identifying the extent of the impact on assets and suggesting measures to restore them to their normal state. Additionally, the analytics can inform future hardening strategies to prevent similar incidents by highlighting vulnerabilities or misconfigurations that were exploited.

Stellar Cyber leverages asset analytics as a foundational element of its security operations and analytics platform, enabling organizations to not only respond to incidents more effectively, but also to proactively manage and mitigate risks associated with their assets. This approach enhances the overall security posture and resilience against future threats.

16:00-17:00

The key components of asset analytics within Stellar Cyber are designed to enhance the security and management of an organization's digital and physical assets. These include:

- **Asset discovery and inventory**. This automatically identifies and catalogs all assets across the network including devices, servers, workstations, applications, and cloud resources. It provides a comprehensive and continuously updated inventory of organizational assets.
- **Vulnerability assessment**. This scans assets for vulnerabilities, misconfigurations, and deviations from best practices or compliance standards. It prioritizes vulnerabilities based on severity and the criticality of the asset to the organization.
- **Configuration monitoring**. This tracks the configurations of assets and detects unauthorized changes or misconfigurations that could expose the organization to security risks.

17:00-18:00

- **Risk scoring**. This assigns risk scores to assets based on their vulnerability assessments, configuration states, and other relevant factors. It helps in prioritizing security efforts towards assets that pose the greatest risk.
- **Behavioral monitoring**. This observes and analyzes the behavior of assets to establish normal operational patterns. It uses this baseline to identify anomalous behavior that could indicate a security threat or operational issue.
- **Forensic analysis**. This provides detailed logs and activity records for assets. It facilitates forensic investigations into security incidents. This helps in understanding the scope of an incident and the path of the attack.
- **Asset criticality analysis**. This evaluates and ranks assets based on their importance to the organization's operations and the potential impact of their compromise. It informs security prioritization and resource allocation.

18:00-19:00

- **Reporting and dashboards**. This offers comprehensive reporting capabilities and customizable dashboards that provide visibility into the security and compliance status of assets, vulnerability exposures, and other critical metrics, and

- **Integration with other security tools**. Asset analytics in Stellar Cyber is part of an integrated security platform, ensuring that insights from asset analytics can be leveraged across other security functions like incident response, user analytics, and network security.

These components collectively empower organizations to maintain a high level of security and compliance posture by providing deep visibility into their asset landscape, identifying and mitigating vulnerabilities and effectively managing asset-related risks.

19:00-20:00
Let's take a moment to check your knowledge.

True or False? Compliance is not a goal of user analytics within Stellar Cyber's platform. The correct answer is false. User analytics can help organizations comply with various regulatory requirements by providing detailed logs and reports of user activities.

What can the integration with other security tools help achieve in Stellar Cyber?

A. Increased marketing capabilities
B. Improved customer service response times
C. Better software development cycles
D. **Enhanced asset management and risk mitigation**

The correct answer is D. Integration with other security tools empowers organizations to maintain a high level of security and compliance by providing deep visibility into their asset landscape, identifying and mitigating vulnerabilities and effectively managing asset-related risks.

20:00-21:00
True or False? User analytics can trigger an automated response to a security threat. The correct answer is true; in some cases user analytics can trigger automated security responses.

The lesson, *User Behavior and Asset Analytics in the Stellar Cyber UI*, is complete. You should now be familiar with Stellar Cyber's user behavior and asset analytics functions as well as their benefits and key components. Thank you!

L05B-DEMO: User Behavior and Asset Analytics (19m:56s)

02K User Behavior and Asset Analytics in the Stellar Cyber UI: DEMO TRANSCRIPT

00:00-01:00
The next lesson in the *Introduction to the Stellar Cyber User Interface* module is *Demonstration of User Behavior and Asset Analytics.* In this lesson, we'll be taking a look at user behavior analytics and how they work in real investigations. We'll explore the value they add to your investigation and the efficiency they provide. We'll also take a look at the asset analytics and show you how to use these as well your investigations and make your investigations as efficient as possible. Let's get started.

Alright. So let's start looking at how user behavior analytics look in the UI. So we'll go to the **Investigate** tab up top user behavior analytics and we get this high-level dashboard that tells us some important information to get us oriented around what users are doing on our network in a given period and you're able to filter either by **absolute**, which is a time period that you assign by just one day or a **relative** time frame...

01:00-02:00
...so we're looking at the last 24 hours here. And you'll notice that there are three tabs here. There's the **Overview** tab, the **Users** tab, and the **Real ID** tab. We'll cover all these. But the **Overview** tab is a high-level view of the most active users on your network. And this might be a way that you orient your workflow to be able to drill down on specific users if you have the manpower to do a workflow like that. So here, by default, we have four charts or visualizations, and starting with the top left we have the **Top Active Users as Source Host**. This chart shows the top users who are actively generating network traffic as the source host, and it provides insights into which users and it provides insights into which users are the most active in terms of sending data across the network.

02:00-03:00
Next, we have the **Top Active Users as Destination Host** chart, and this displays the top users based on their activity as destination host in the network. It shows the users who have received the most traffic or connections from other hosts in the given time frame. The **Top Users with Alerts** chart shows the users who have the highest number of Alerts triggered. It provides a ranking of users based on the number of Alerts associated with their activities. So this might be a chart that you **use in your workflow** to investigate which users are generating the most Alerts, and we're going to take a look at this user Rossan in just a minute. So if you have the manpower to do investigations about which users are creating the most Alerts on your system, this would be a good way to create queries, or to go filter in the Cases and start with Cases or go look at their Alerts to orient your workflow around users.

03:00-04:00
The **Top 10 Users by Risk** chart queries Stellar Cyber's Mongo database for the top-ten users that have the highest risk scores. And just like the **Top Users with Alerts**, this might be a chart that you use to go query and say I want to take a look at this user and figure out what he's doing. So you might pivot over into their Cases, take a look and see if they have any. We'll do that now. And we'll filter by this name. And we do have a Case. So if we click into this Case, we can see that this user is involved in this Case. We'll find their name here. So we see the target username is Rossan. So we've got the person that we were looking for. So those are some ways that you can use these charts to orient your workflows.

04:00-05:00
Also we have the ability to edit these charts. If we would rather have the top users up top, we can just click and drag these where we want them. And get these out of the way if we don't care as much about these or we just want to see these first. We can generate a preview of this to see what it's going to look like. And it lays it out for you nicely. And then we can save that if we'd like to. And then it tells us automatically in a big red box that you're not looking at the default menu. You're not looking

at the default dashboard. So we're going to restore the default here. And that's how you edit the layout of this dashboard.

Also you could clone this dashboard; give it a custom name, give it a description, and when you clone it, it's going to save it as you have it filtered. So if you wanted to look at specific types of data by filters, you could filter the data in here and you could look at this dashboard in the way that you want to see it when you get started every day.

05:00-06:00
And then there's an export that if you click here you can save this to a PDF. And here you have a PDF format that you could download and share with other security stakeholders or other members of your team as a daily snapshot of what you're looking at workload wise in your SOC.

On the **Overview** page you can see the filters work in the same way as the Cases and Alerts page. It is filtered by Alerts score if that's the way you want to see it. You could choose a prebuilt query. You can choose **Status** or add a new filter Any of these options are a filter and you can add more than one filter at a time. And this works across all of the user behavior analytics views.

06:00-07:00
And one last thing to note about the Overview here is that you do have these filters, if you wanted to clone your dashboard, with the filters. You can filter here by Alert score. You can add a custom query that you've saved. You can filter by status or add a new filter. Any of these fields can be filtered on. This is a way to get to the data that you want to see quickly. So the filters work the same way across views. So we'll hop over to the user's view. And honestly, this is where you're going to do most of your work as far as triage and investigation from a user-centric perspective. And we'll go ahead and auto size all these so that we can see all of the data in these fields. And the filters work the same way with the difference being you have a lot more options here that you can filter on any of these. You could filter on description, you could filter on location, and all of these are selected by default. You can see it here, but the filters work in the same way across views as do the column choices.

07:00-08:00
So you can choose to have more fields in your data table, than the ones that appear by default. You can pick any of these or you can add them from the menu here. So if you want to add this column, you can look at any of these fields here, and add this column to your view. So we'll add Tenant ID just as an example, but it works the same way across all the views.

So let's talk a little bit about how you actually conduct your triage investigation in this view. So you can see here that there's not a field that shows you how many Alerts are on this user. We can see, however, that the risk score is very high, and we can sort them in this way and try to work through each one of these by their risk score. So first you can see some important data about the user and their user history.

08:00-09:00

This person has got a high-risk score and lots of them, but this just gives you a little bit of information about the user themselves. And to see the data about this user, if we want to investigate this, we'll click into the user. So in your environment there will probably be a lot more data populated. This data comes from Active Directory and gives you critical contextual information about this user, including their name, principal name, the SID if you want to search that way, their job title, department, and manager, as well as the company name if you have more than one company in your instance. But the job title information gives you critical information in that you can add key context to an incident or an event or an Alert by looking at their job title, looking at the activity, and seeing if there might be a valid business operation that requires this type of activities.

09:00-10:00
So if it's port scanning going on, you might look to see if this person is a system or network administrator; that might be normal behavior for them. If there's network enumeration going on and it's a human resources employee, that's strange behavior. And it definitely is going to increase the criticality of the event. So this information is crucial, and you'll find it here. You also have some useful contact information here. So if there is activity that's going on like maybe you get an Alert that someone has used some sort of file transfer protocol that's against the policies of your organization, it may be if you look at the activity it's okay to contact the user directly and say, "Hey did you download some sort of FTP agent? And if so, what are you using it for?" to explain the activity. If the activity looks like it might be insider-threat related or it might be a malicious actor having access to your network, you may want to call the manager and say, "Is this okay for this user to engage in this type of activity on the network?" Because it may be a special use case that they have to do so, and you just have to judge that by your investigation and how it goes.

10:00-11:00
So we'll scroll down and look at what else we have down here. We have some useful metrics here--latest risk score, number of Alerts and that critical. That's what we were kind of looking for in the original view to assess how many Alerts this user is generating. This is a lot of Alerts for one user. Also we have authentications, and certain applications will authenticate throughout the course of the day. You may see lots of authentications that don't really pertain to what you're investigating, but this information is useful to have, and you can find it here in the Authentications tab.

So best practice would be to go into Cases and find this user. So what we'll do is we'll search for the username and see which Cases are involved if any. So we have this Case here, sunburst, which is a known type of attack.

11:00-12:00
We'll click into it and see that there are thirteen Alerts here, 13 of a 113, so this will at least knock out 13 at once. So if we go in here and look at the timeline to assess whether this is a critical Alert and we need to maybe isolate this user, block their access, we can take an assessment of it really quickly here. So we have a brute force successful user login: clearly bad. Login time anomaly: also bad, port scan activity, reconnaissance, some sort of exploit, domain-generated algorithm involved, merging threat, trojans. We have all this. So now we have really enough information to assess whether this

user and we can click back into the user details using the **user detail view** here, we just click on the user and click back in the **user detail view** to take us back to the **user behavior analytics** field.

12:00-13:00
This is possible with an Alert as well and we'll explore that later on in this lesson, but this gets you back to do investigation on what types of Alerts and activities this user has engaged in. It gives you critical insight into what this user does habitually. So we've got lots of Cases where we have incidents for this user across the month of March here. Lots of login failures. That's a sign and we've already seen that there was a brute-force attack happening here. So this is all critical information for you. So that's how you interact with user behavior analytics through Cases.

So let's take a look at how we investigate the Alerts for this user. If we go down here, we have all the Alerts already here. But if we happen to be in a list of Alerts with different users what we can do is go in here and find the source username for this user and create a query.

13:00-14:00
So that if we wanted, we could go into the **Alerts** page, and copy this query, click **Enter**, and we have all of these Alerts. They are grouped by type. But the best practice would be just to investigate it in the user behavior analytics for the user. And we can investigate these here. Also in this view, we can view related assets for this user. Here we have lots of authentications. We have the source IP for this user. We have commands run for this user. This doesn't involve commands. We have these many authentications we can view here, source IP, source host, destination IP if it's there.

14:00-15:00
Then we can see the login failures. And any files involved in these Alerts. That's how you pivot back and forth between Alerts and Cases and the User ID.

The purpose of the Real ID tab and user behavior analytics is to display users with their various identities. It shows a table of users with all of their different identities imported from Active Directory and correlated with a single SID. So if we filter on Rossan we'll see that there are two Alerts, there's a domain account and there is a username and we can investigate these accordingly.

Alright. Let's switch gears and start looking at the asset analytics functions that we have here in Stellar Cyber.
15:00-16:00
So your first tab here is a Dashboard that gives you valuable information about the vulnerabilities and the risk on your network. So it's a premade dashboard that shows you valuable information about your assets, like the risk distribution, vulnerable assets, this may be something you want to immediately take care of. This asset here has lots of vulnerabilities that might be exploited. It will show you unauthorized assets on your network if you have any, and then Asset status. So there are five not active on the network. It shows you Asset Approval, Asset Value, Asset types whether they be clients or routers or servers asset tags. If you had asset tags in your instance they would appear here. And then you have Asset Locations, as well as your top asset vendors So there are 138 vendors that are using VMware here. And then we'll move over to the MAC-identified assets.

16:00-17:00
Here you have a view that is much like the user's view in the user behavior analytics. And from here you could do your investigations on the most risky assets in your environment. So in this scenario we'll say we don't really have the workforce to evaluate all of these every day. So we will scale this up to 80. In this particular case, it may not be every case, but we will take a look at these and we have 19 results to triage. So we'll just pick one at random. Here, the Asset Details View gives you the friendly name, which is the IP in this case, the ID, your tenant, the importance, reputation, description if there is one, your first and last scene. So if you have new assets on your network you can evaluate that here, the location, any tags that exist on this asset, the type of asset it is, the sensor that is collecting this information in the VMware, and the operating system, if you happen to have one.

17:00-18:00
So Latest MACs this one only has one, latest host names. This one has a few that we can evaluate in the Panorama view. Latest IPs. We could also evaluate that in the panoramic view if we wish to. And the types of applications that this asset is accessing. So this can give you valuable information about what sort of protocols this asset is seen using and when. And here is where the value is. These are all of the Alerts that occurred from this source host. So we could evaluate one at a time or we can go in and see if this belongs to a Case. So this one does happen to be included in a Case. So we'll pivot into the Case and take care of all of these Alerts at one time.

18:00-19:00
So we have a lot of persistent foothold Alerts here. We can look at this asset and oh guess who this is? Rossan. So if we wanted to pivot back into his user details we could do this here. We can look at the IPs that it's connecting to. And here we have the executables involved in this investigation. The timeline here we have uncommon process anomalies, RDP registry modification, all this is bad stuff and we can see it all in one view. So we might want to go ahead and isolate this host because instantly we can tell that this is a dangerous asset. So we'll go in and perform an investigation just as we would on a user, but this time we'll be looking at the assets. And here we have the host processes that were involved,

19:00-20:00
registry it's on, and the user. So under the **MAC-Identified Assets** tab you could investigate these one at a time or you could look for them being aggregated in Cases, and that's basically how you investigate the risky assets on your network.

Likewise with the IP-identified assets you would just click through here and investigate these one at a time. You've got your routers here, information about routers, asset imports if you have any, and their associated certificates. So that's essentially how you do investigations in the asset analytics and how they interact with other elements within the UI. So that concludes this lesson. Happy hunting. This lesson is complete. Thank you for watching.